Matching Spirits Supply to Demand in Iowa

Decision support for the Iowa Alcoholic Beverages Division (ABD)

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Stage 1 · Define the problem

Stakeholder · questions and hypotheses · data audit · integration plan · project plan

ABD must commit stock before stores order

Stakeholder

Iowa Alcoholic Beverages Division: sole spirits wholesaler. Buys from suppliers, holds one central warehouse, delivers to every licensed store and issues their licences.

Decision

How many liters of each spirits family to buy and truck to each county next month, and how to treat border counties and new licensees.

KPI and constraints

Cut planning error (WAPE, county × family) by ≥10% vs “same month last year”, bias within ±3%; translate into safety stock. Public data only · one state price · ABD is also the regulator.

≈2,240

licensed retailers supplied by ABD

4.5M

order lines, Jan 2025 – Oct 2026

−3%

statewide liters, Jan–Oct 2026 vs 2025: last year's plan over-buys

Three pre-registered hypotheses, each tied to a decision

H1 · Cross-border leakage

Counties where cheaper out-of-state spirits are close buy fewer liters per adult (Missouri tax $2.00/gal vs Iowa $15.14).

Decision: per-capita planning norms; state revenue.

H2 · Regional mix

Counties with a larger Hispanic share buy a larger tequila/mezcal share.

Decision: category mix supplied to each delivery region.

H3 · Store entry

A new licence takes volume from existing stores, so the city grows by less than the newcomer sells.

Decision: how to plan and license new stores.

Written before any 2026 outcome was examined: 2025 = discovery sample, Jan–Oct 2026 = confirmation sample. This fixes the pre-submission caveat (patterns were “confirmed” on half of the same year).

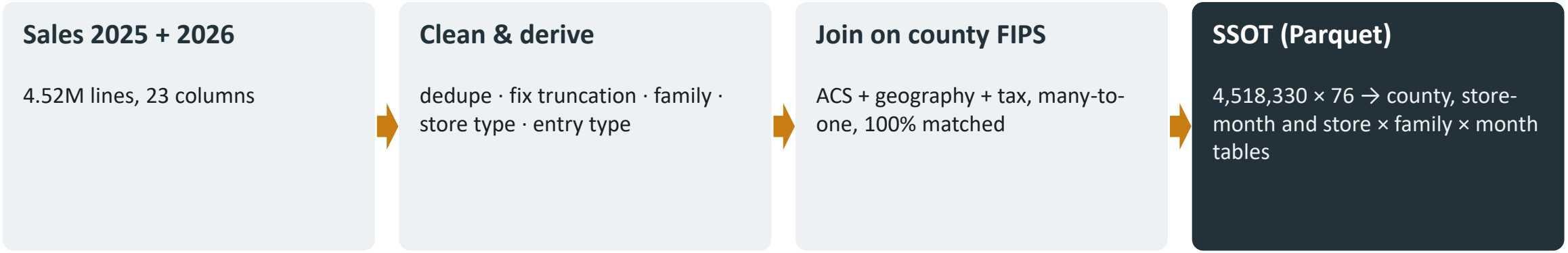
Dropped from the pre-submission: income → premium share (rejected), young adults → RTD (a proxy for urbanisation), K-means (not in the course handout).

Data audit: five official sources from four publishers

Source	Publisher	Coverage	Size	Role
Iowa Liquor Sales 2025 (dataset 1262)	Iowa Data Hub (ABD / Alcohol & Tax Ops)	Jan–Dec 2025	2,490,620 rows	orders · discovery sample
Iowa Liquor Sales 2026 (dataset 1263)	Iowa Data Hub	Jan 1 – Oct 30 2026	2,029,764 rows	orders · confirmation and test
ACS 5-year 2020–24, 16 tables	US Census Bureau	99 counties	22 variables	demographics
County adjacency + Gazetteer 2024	US Census Bureau	99 counties	15 variables	borders, area, distance
State spirits excise, Jan 2026	Tax Foundation (cross-check: FTA)	Iowa + 6 neighbours	7 rows	tax gap for H1

Row counts match the publisher's API exactly for both years · no Kaggle · licences and links in data/links.txt · access: open API, no constraints

Integration: one source of truth, 4.52M rows × 76 variables



- Join keys are codes, not names: 5-digit county FIPS (federal standard); store_no (stable across years, 2,078 stores in both)
- Grain: one row per order line; aggregated to county (hypotheses) and store × family × month (forecast)
- Units: liters recomputed from bottles × bottle size; US dollars; every step logged in data_quality_report.json
- Reproducible: scripts/01_build_ssot.py rebuilds everything from raw files in about a minute

Plan: BADIR mapped to course methods

BADIR	What we do	Course methods (handout unit)
Business question	ABD's monthly supply and licensing decisions; KPI = WAPE and bias	Problem framing, KPI and metric design
Analysis plan	Pre-registered H1–H3; time-ordered split; status-quo baseline	Hypothesis testing; experimentation design
Data	ETL into an SSOT with quality gates	ETL pipelines, data warehousing
Insights	OLS + permutation + bootstrap; stacked difference-in-differences	Hypothesis testing; causal inference (DiD)
Recommendations	Gradient-boosting ensemble, calibrated intervals, newsvendor plan; API, MLflow, Docker, drift; SHAP, fairness	Feature engineering, ensembles, boosting, HPO, calibration; optimisation; MLOps; responsible AI

Not forced: Spark (data fit in memory), deep learning (tabular monthly data), RL (no sequential control), A/B test (proposed for the roll-out instead), K-means and LIME (not needed).

Stage 2 · Validate the evidence

Data preparation · EDA · hypothesis tests · inconclusive results · 20 more hypotheses · revised analysis

Data preparation: four defects the pre-submission missed

Truncation in three months, not two

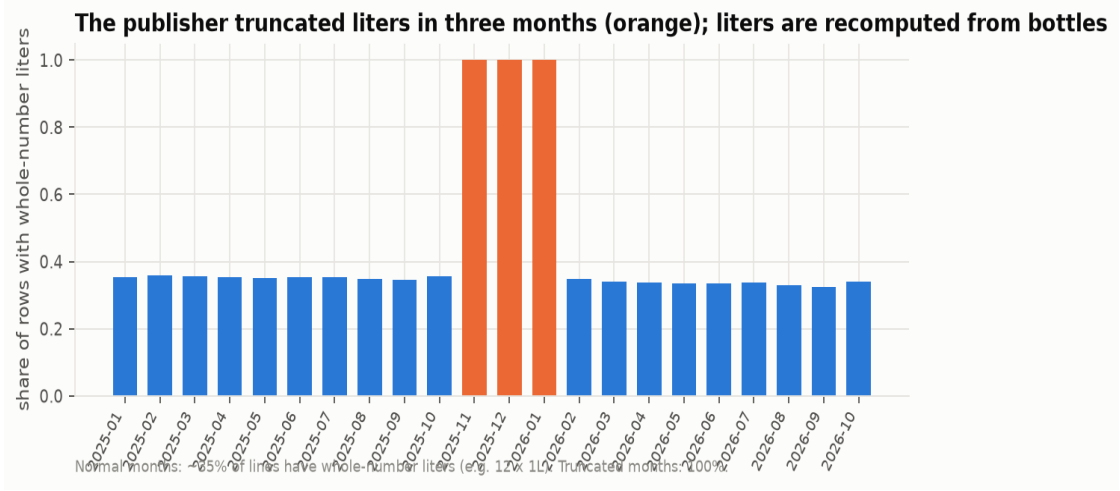
Liters, dollars and prices are whole numbers in Nov–Dec 2025 and Jan 2026 (liters −3.4%). Liters recomputed; dollars rebuilt from item prices (validated: −0.08% error).

Multi-pack minis

313 “50 ml” items are really sleeves or buckets of minis (e.g. $81 = $1,620/L). Flagged; excluded from price metrics; ~2% of liters understated; H1 re-tested in dollars.

Ownership changes ≠ new stores

55 “new” licences sit at the address of a store that just closed. Separated from 281 genuinely new stores; otherwise H3 would be wrong.



Also handled: 429,989-row duplicated download part · 21 exact duplicates · 4,135 blank counties (2,102 imputed from the store) · 4 renamed categories · returns kept as negative volume.

EDA: three patterns that change how ABD should plan

−3%

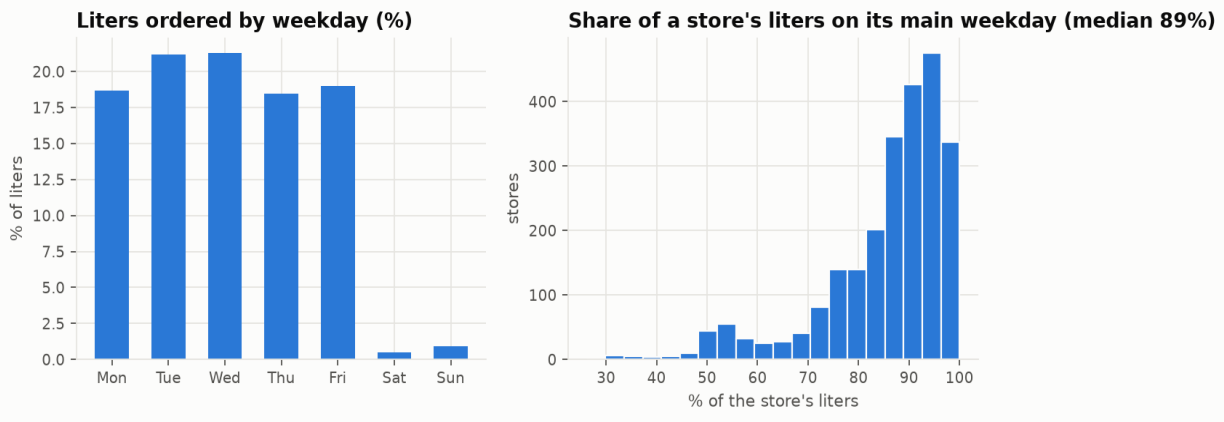
2026 vs 2025, nearly every month → last year's plan systematically over-buys

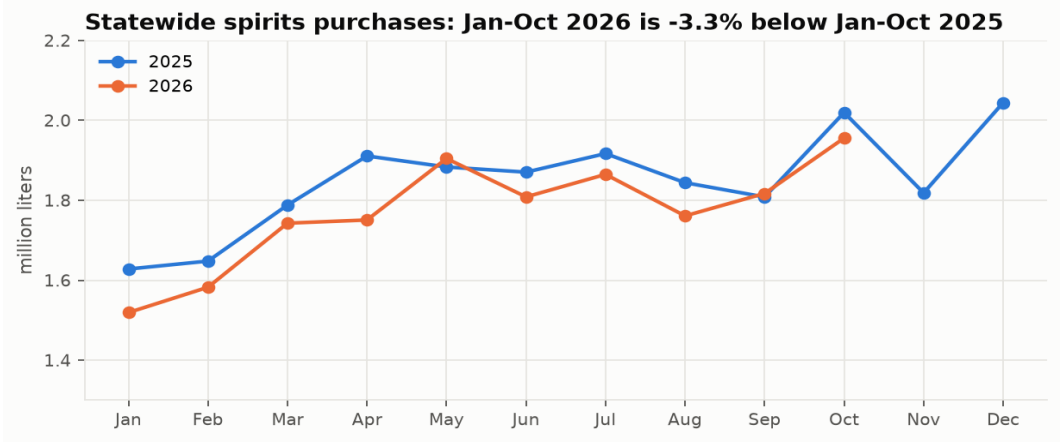
89%

of the median store's liters arrive on one weekday → 4 vs 5 order days = 25% swing

½

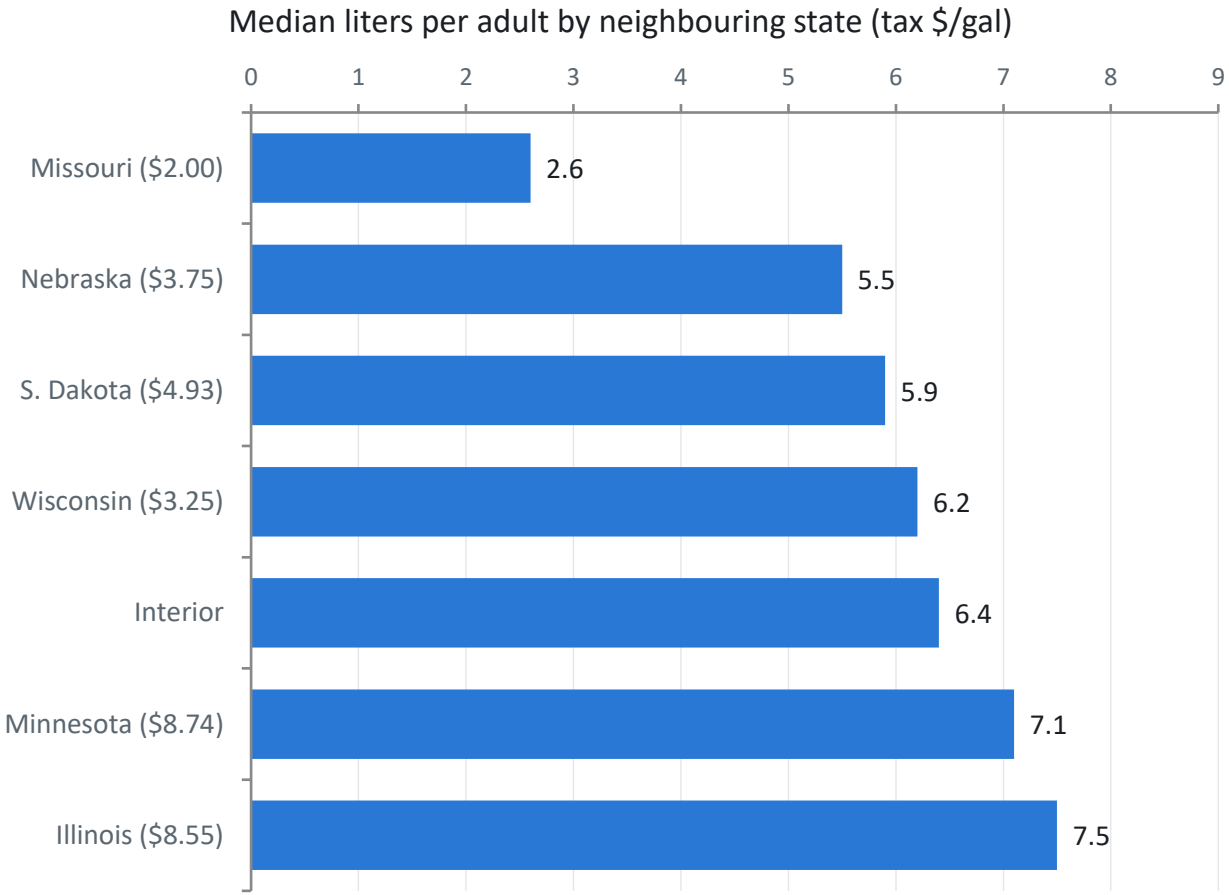
liters per adult in Missouri-border counties → per-capita norms fail there





Observation → inference → decision: drift becomes the baseline to beat; the order calendar becomes a model feature; the border becomes H1.

H1: Missouri-border counties buy 44% fewer liters per adult

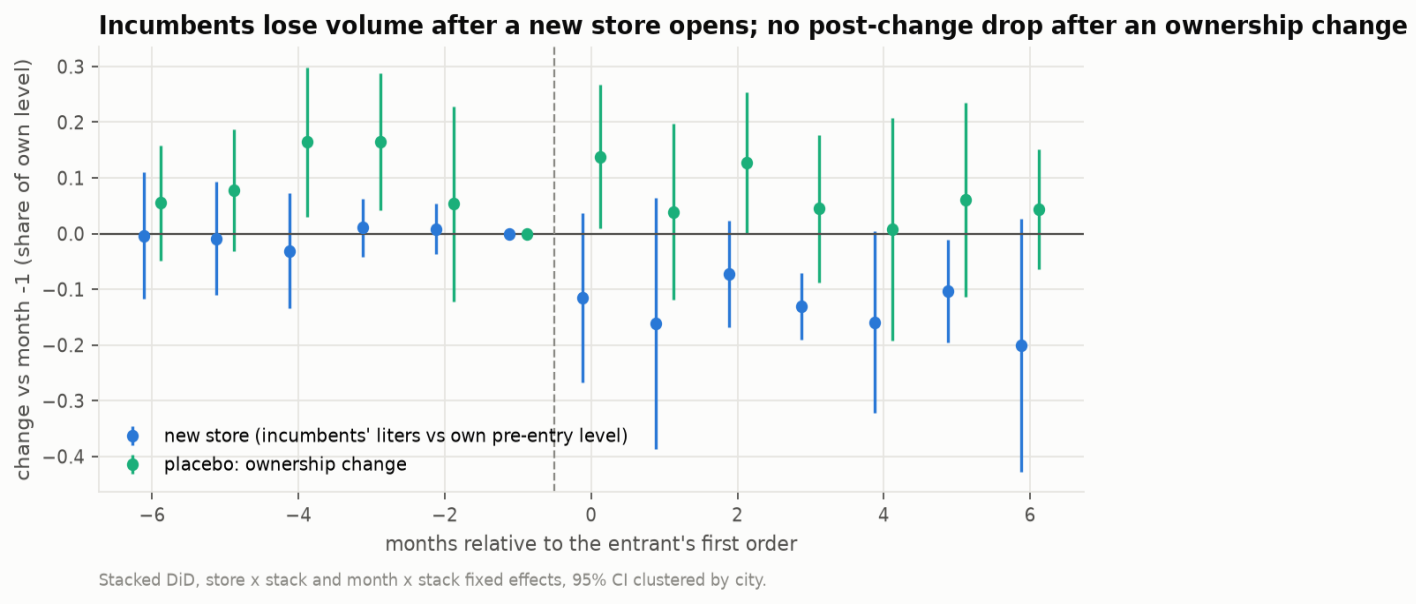


−44%

liters per adult vs comparable counties, 2026 (95% CI 25–58%); 2025: −43%

- Controls: population, income, median age, education
- Passes all four checks: HC3, permutation, trimmed, population-weighted; also holds in dollars
- Shortfall fades with distance from the border in both years
- At stake: ≈$5.9M sales ≈ $2.0M ABD margin a year (upper bound; association)

H3: a new store mostly takes sales from existing ones



−13%

incumbents' volume after a genuinely new store opens (p = 0.01, flat pre-trend)

≈60%

of the newcomer's volume diverted from incumbents (95% CI 28–129%)

- Stacked difference-in-differences: 116 cities, 1,136 incumbents, 475 city clusters
- Small incumbents −21% vs large −4%; ownership changes show no post-change drop
- H2 also confirmed out-of-time: +0.8 pts agave share per +10 pts Hispanic population

Inconclusive and unsupported results are findings too

Result	What the evidence says	Status
H1b tax-gap dose-response	Border groups line up with the tax gap; significant in 3 of 4 checks, fails population weighting (p = 0.21)	Inconclusive
H1c fewer 1.75 L “stock-up” bottles	−7 pts at the Missouri border; fails population weighting (p = 0.23)	Inconclusive
H1d leakage to a bigger Iowa neighbour	p ≈ 0.10 once age and education are controlled	Not supported
Pre-sub H3 income → premium; H4 age → RTD	Rejected / confounded by urbanisation	Dropped, reported
H3 raw-liters outcome	Not significant: a few giant stores dominate the variance; normalised outcome used	Disclosed deviation
H3 placebo	No post-change effect, but a noisy pre-period (20 cities)	Weaker than hoped

Twenty more hypotheses: 10 supported, 3 partial, 7 rejected

Geography & culture

H4 Brandy belt crosses the river
● Supported

H5 Taste spreads to neighbours
● Supported · Whiskey, rum

H6 Missouri pull ends at ~30 mi
● Partial

H7 Sioux County is a 'dry' outlier
● Rejected

H23 Whiskey divide is generational
● Rejected · Rural, not age

Cultural calendar

H8 St Patrick's lifts Irish whiskey
● Supported

H9 Cinco bump is Hispanic-led
● Rejected · Statewide

H10 College towns empty in summer
● Supported

H11 Holiday trade-up is for the rich
● Rejected · All incomes

H12 Palate rotates with seasons
● Supported · 7 of 8

Prices & economics

H13 Price cuts: stock up, then dip
● Rejected · Reversed

H14 Tariffs raised import costs
● Rejected

H15 Inequality builds a barbell
● Partial

H16 50 ml nips track poverty
● Partial

H17 More outlets, more drinking
● Supported · Under 1:1

Market & stores

H18 The long tail lives in cities
● Supported

H19 Spirits follow Zipf's law
● Supported

H20 Opening-order honeymoon
● Supported

H21 Dying stores fade slowly
● Rejected · Sudden

H22 Customers stay with the address
● Supported

Exploratory, not pre-registered. County patterns must hold in both 2025 and 2026. Six survive a Holm correction for 20 tests: H6, H12, H18, H20, H21, H22.

Geography: tastes cross borders, and age is a red herring

H4	**Wisconsin's brandy belt crosses the Mississippi** Border counties 2.7% brandy vs 1.6% elsewhere; holds 2025 and 2026	Supported
H5	**Taste spreads to neighbouring counties** Whiskey and rum residuals cluster (Moran's I 0.14-0.17); agave and gin don't	Whiskey, rum
H6	**Missouri's pull ends at a travel horizon** 2026: +34% liters/adult per 10 mi up to ~30 mi, then flat; 2025 linear	Partial
H7	**Dutch-Reformed Sioux County is a 'dry' outlier** Ranks 40 of 99 (−7%). Real outliers: Louisa and Keokuk	Rejected
H23	**Canadian = old Iowa, bourbon = young Iowa** Raw split looks generational; with controls the age effect flips	Proxy

1.7×

brandy share in counties on the Wisconsin border (H4)

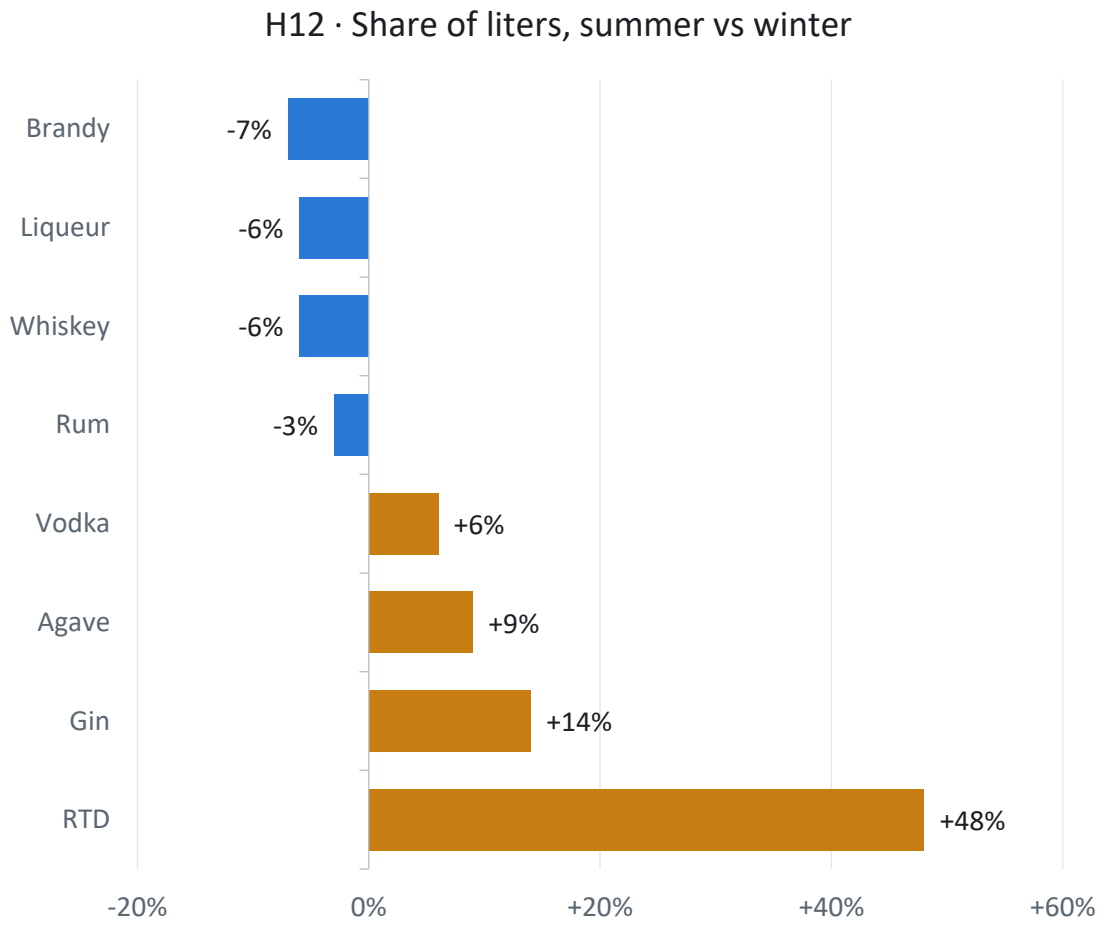
~30 mi

how far Missouri's cheaper spirits pull Iowa buyers, 2026 (H6)

−21 pts

Canadian share of whiskey per 10× density: it's rurality, not age (H23)

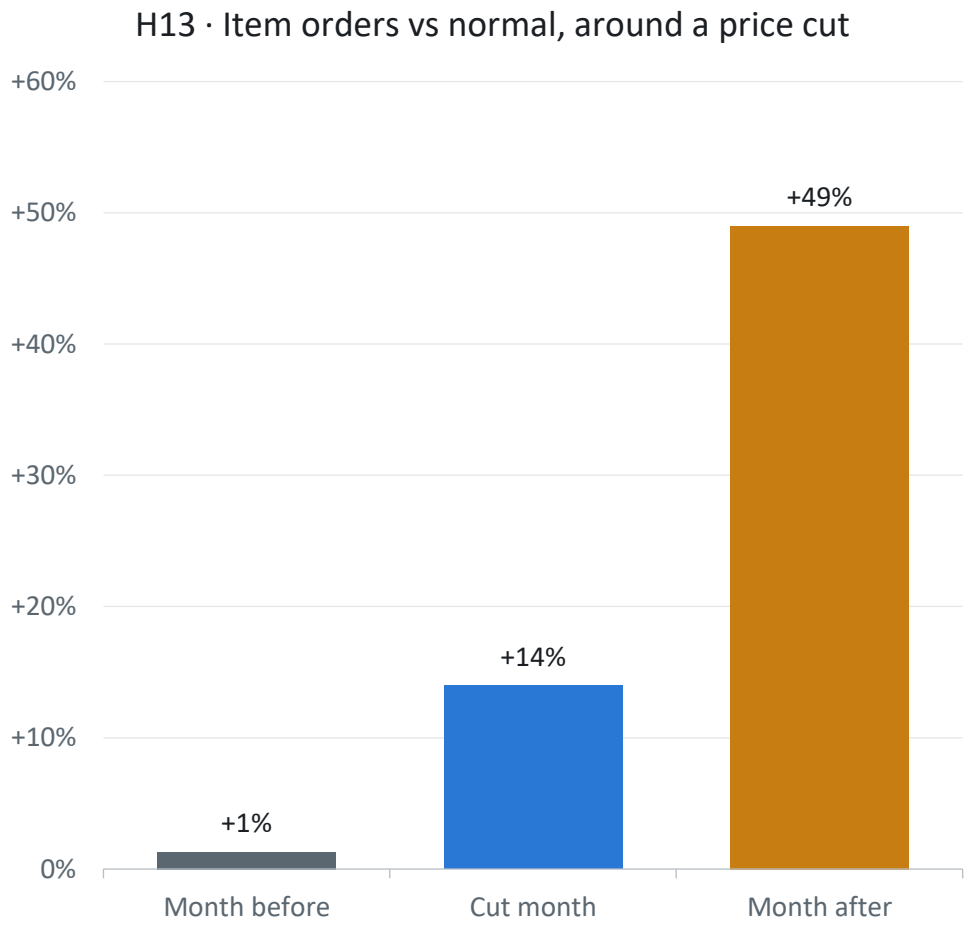
The calendar: restocking runs weeks ahead of every occasion



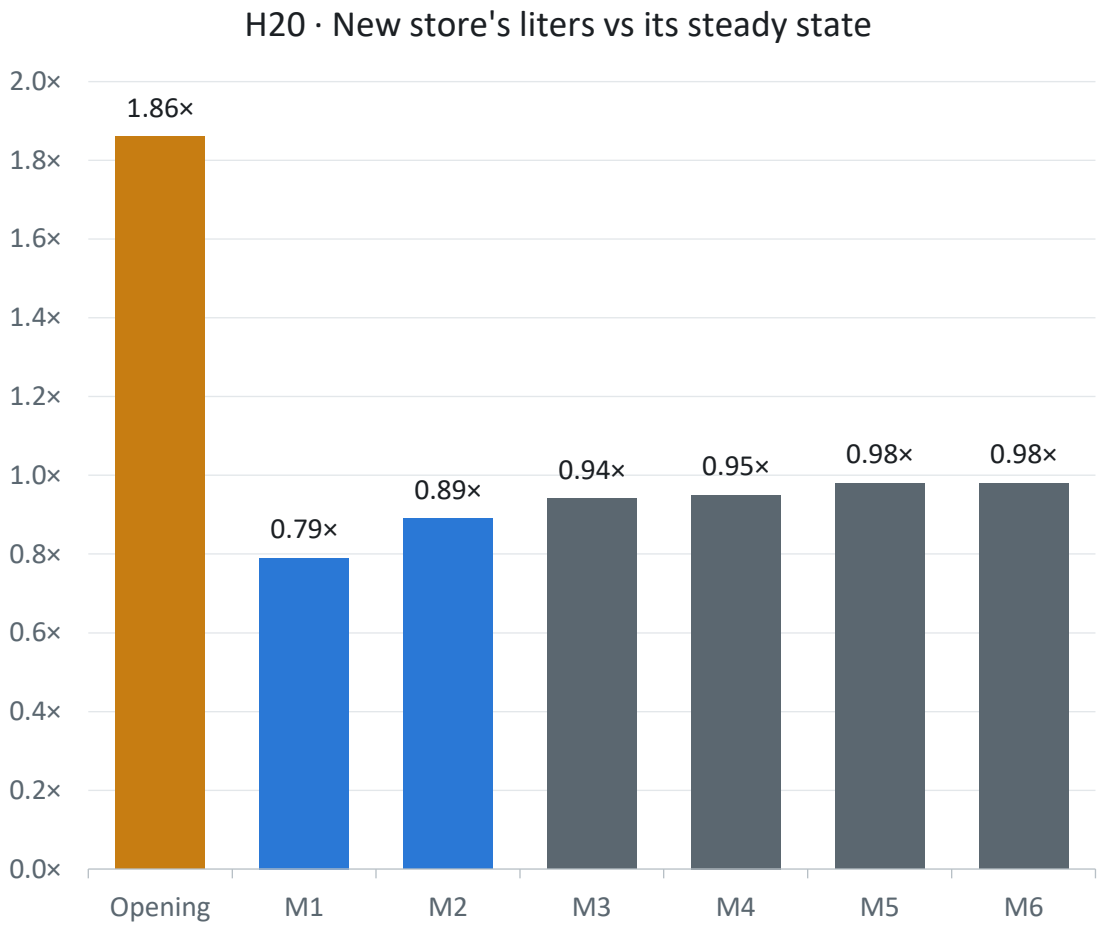
H8	**St Patrick's Day lifts Irish whiskey** +29% share in Feb-Mar (2026), +24% (2025); Scotch +4%	Supported
H9	**The Cinco de Mayo bump is Hispanic-led** Bump is real (+16%) but statewide: a party, not an ethnic, occasion	Rejected
H10	**College towns empty out in summer** Ames −37%, Iowa City −25% summer vs autumn: 2nd and 4th lowest of 99	Supported
H11	**Holiday trade-up is a rich-county thing** Premium share 24.8% → 27.1% in Nov-Dec, equally at every income	Rejected
H12	**Iowa rotates its palate with the seasons** 7 of 8 as predicted. Surprise: rum is a winter spirit	7 of 8

Prices: price cuts pay off late; new outlets add demand

H13	**Price cuts: stock up now, under-order later** Reversed: flat during the cut, +49% the month after; no dip	Reversed
H14	**Tariff era made imports dearer** Same-item cost moved under 1% for every origin, Jul-Oct 2026 vs 2025	Rejected
H15	**Inequality builds a price barbell** High-Gini counties buy a wider price spread, but pricier on average too	Partial
H16	**The 50 ml nip is a poverty index** No county effect; within a store type +3.3 pts per +10 pts poverty	Partial
H17	**More outlets, but under 1:1 more drinking** Elasticity 0.39-0.52 (pop.-weighted) vs 0.40 implied by H3	Supported



Stores: a big first order, a sudden exit, and loyal addresses



H20 — **New stores have an opening honeymoon**
Opening order 1.86× steady state, then 0.79×; steady by month 5
Supported

H21 — **Dying stores fade slowly before exit**
Only −6% before the last order; warning AUC 0.53 (coin flip)
Sudden

H22 — **Customers belong to the address**
New owner's mix 2× closer to predecessor's; keeps 92% of volume
Supported

H18 — **The long tail lives in cities**
Same order lines: 547 distinct products in dense vs 322 in sparse counties
Supported

H19 — **Spirits follow Zipf's law**
Slope −1.10; top 1% of 4,792 products = 38% of liters
Supported

Five findings to take into planning

H13

+49%

orders the month after a price cut ends

Decision
Plan promotion stock for month t+1, not t

H17 + H3

~45%

of a new outlet's volume is new demand

Decision
Two methods agree; net entrants in capacity plans

H22

92%

of volume stays when a shop changes hands

Decision
Forecast from the predecessor's history

H20

1.86×

a new store's opening order vs normal

Decision
Cap entrants at month 0; use peers until month 3

H23

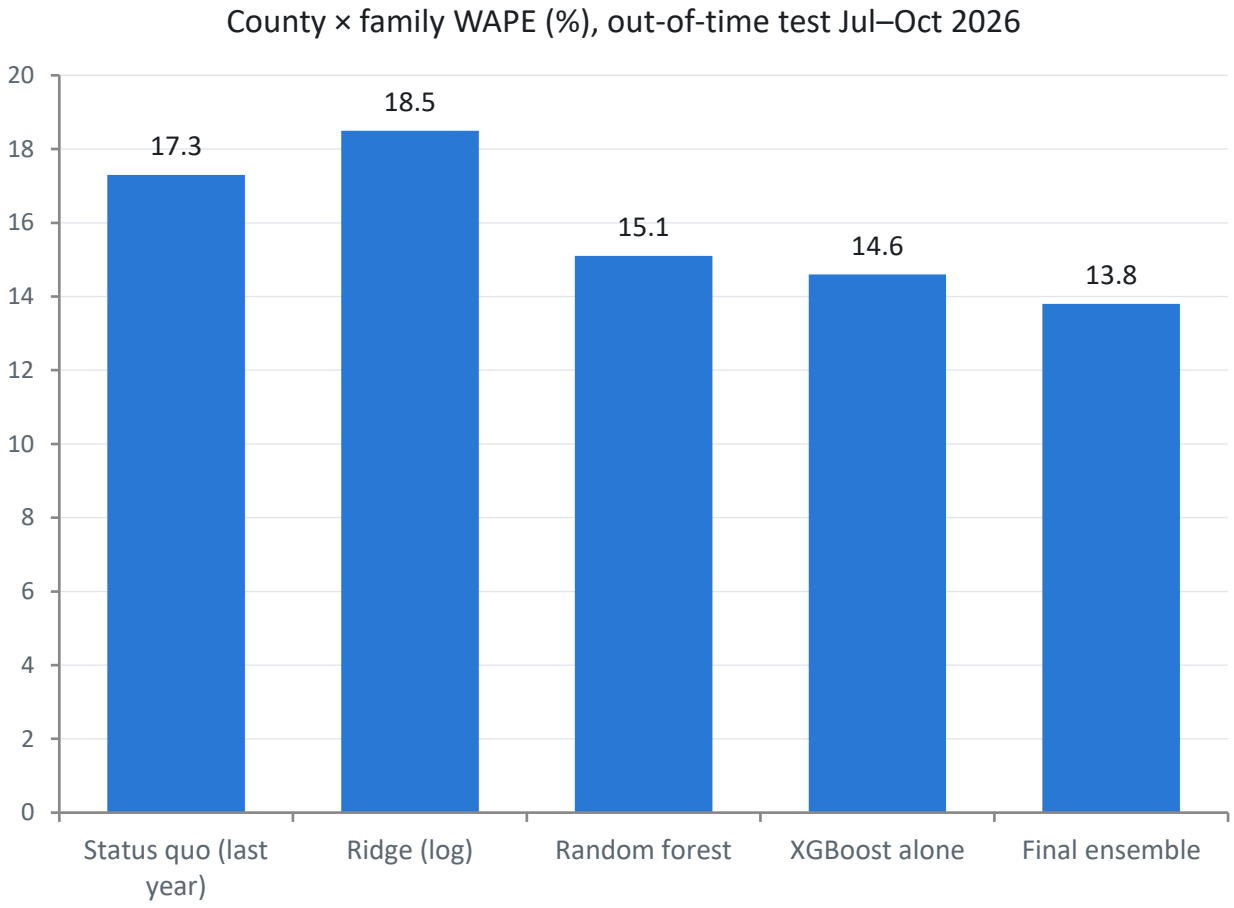
−21 pts

Canadian share per 10× density

Decision
Allocate whiskey by rurality and education, not age

Full tests, verdicts and caveats: reports/twenty_hypotheses.md · scripts/05_twenty_hypotheses.py

Revised analysis: from explaining demand to forecasting it

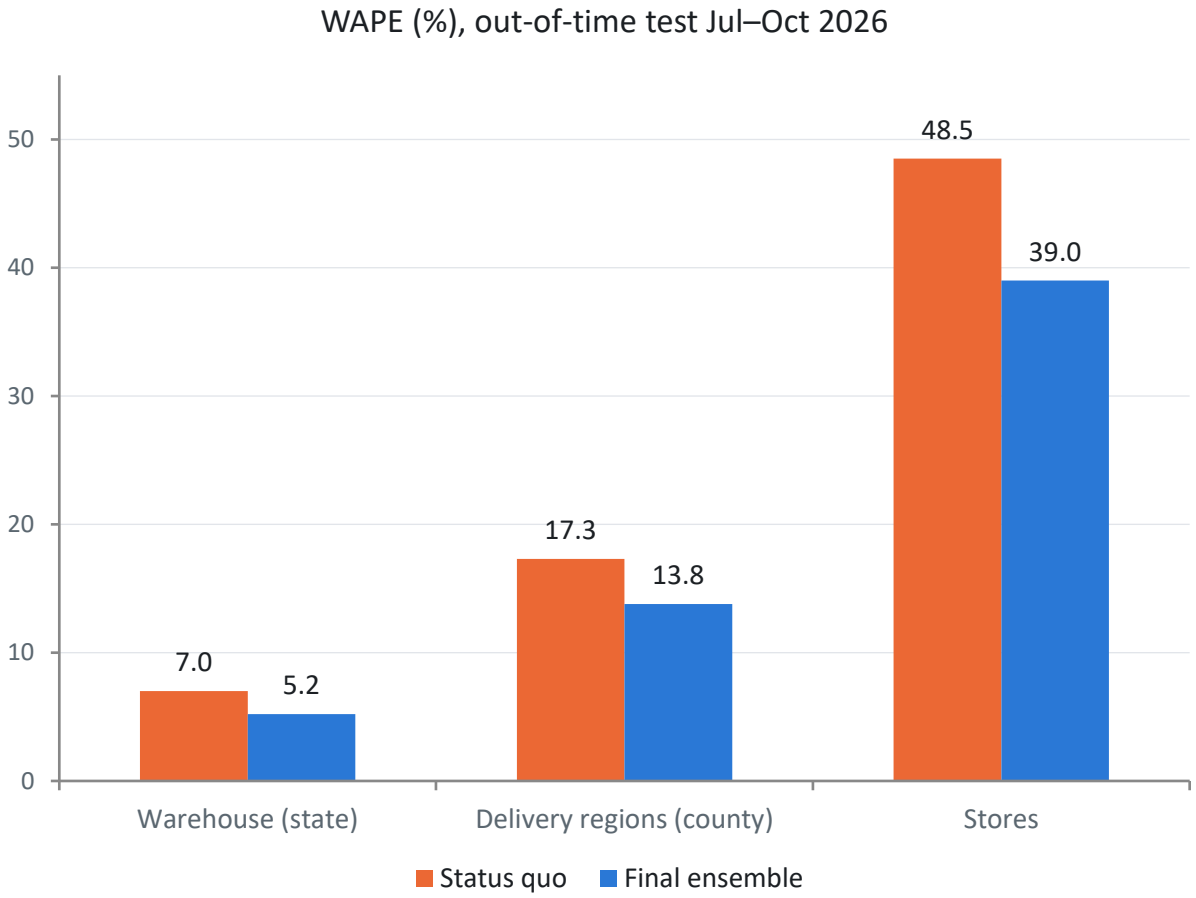


- Unit: store × family × month (409K rows); features only from data known the month before
- Learned models under-forecast (one seasonal cycle; trees can't extrapolate giant stores); last year over-forecasts in a shrinking market
- Opposite errors → ensemble: 0.55 × Tweedie XGBoost + 0.45 × calendar-adjusted seasonal naive (weight chosen on validation)
- HPO (20 trials) changed little: gains come from features and design

Stage 3 · Inform the decision

Final findings · recommendations · impact and limits · roadmap

Final findings: 20–27% less error and 23 hypotheses tested



	Finding	Status
H1	Missouri border: −44% liters per adult	Supported
H2	Agave share rises with % Hispanic	Supported
H3	New licences: −13% incumbents, ~60% diversion	Supported*
H4–H23	20 exploratory tests: 3 partial, 7 rejected (slides 14–19)	10 supported
Model	Bias +2% (status quo +6.5%); calibrated P10–P90	Meets KPI

Strongest of the 20

· Price-cut orders arrive a month late: +49% (H13)

· ~45% of a new outlet's volume is new demand (H17 + H3)

· A change of owner keeps 92% of a shop's volume (H22)

* with one disclosed deviation (normalised outcome). H1b, H1c inconclusive; H1d not supported. H4–H23 are exploratory, not pre-registered.

Recommendations for ABD

1. **Plan procurement and capacity with the ensemble forecast; stock price-cut items for the month after**

   −20% / −27% error · H13: +49% a month late

2. **Set stock by service level, chosen per family from stock-out vs holding cost**

   −25% safety stock

3. **Plan delivery capacity by order days, not months; stock occasions and college terms weeks ahead**

   89% on one weekday · H8, H10, H12

4. **Drop per-capita norms in a ~30-mile southern band; plan those counties on their own history**

   H1: −44% · H6: ~30 mi

5. **Brief the board and legislature on the Missouri border gap (≈$2.0M margin/yr, upper bound)**

   H1 + tax data

6. **Forecast new licensees from peers, ownership changes from the predecessor; trim nearby incumbents**

   H3: ~60% diverted · H20: 1.86× opening · H22: 92% kept

7. **In licence reviews, report the hit to small incumbents and the ~45% net-new demand**

   H3: −21% small · H17: elasticity ≈ 0.4–0.5

8. **Set regional mix from local drivers (agave, NE brandy, rural Canadian); keep ethnicity out of the model**

   H2, H4, H23; ablation

9. **Ask the publisher to fix truncated months, blank 2025 costs and mini pack counts**

   data audit · H13

10. **Roll out via a randomised regional pilot (A/B) before going statewide**

    causal proof of value

Impact, responsible AI and limitations

−25%

warehouse safety stock at 95% service ≈ $0.73M working capital

84K L

per month of systematic over-buying removed (≈$1.0M at cost)

0

personal data · ethnicity dropped from the model (no validation gain)

Responsible AI

SHAP: own history 78%, store 11%, demographics ~1%
Fairness: better than status quo in every group, but +7% bias in rural / border counties vs +1% urban; new stores −24%
Monitoring: PSI, monthly error, schema drift, retrain rule; one false alarm diagnosed

Limitations and assumptions

22 months = one seasonal cycle
Sales recorded at the store; census 2020–24; ecological data
H1 association; H3 CI wide; revenue figure an upper bound
Value assumes 1-month lead time, 20% holding cost

Implementation roadmap

0–2 mo	2–5 mo	5–8 mo	3–9 mo	Ongoing
Shadow mode	**Randomised pilot**	**Statewide roll-out**	**Policy use**	**Run & improve**
Analytics team runs forecasts beside the current plan	Half of delivery regions plan with forecast + service levels	Procurement by service level; order-day capacity plan	H3 note in licence reviews; border-gap briefing (H1)	Monthly refit, drift and fairness audit, publisher data fixes
KPI: county WAPE ≤ 14%, |bias| ≤ 3%	*KPI: fill rate ≥ today; inventory days −10%*	*KPI: safety stock −20% at equal fill rate*	*KPI: briefing delivered*	*KPI: no group |bias| > 5%*

Owners: ABD supply-chain analytics (forecast, monitoring) · ABD planning and operations (pilot, roll-out) · ABD licensing and board (policy use)

Demo and decision challenge

Live: FastAPI planner at localhost:8000 · county plans, service-level slider, store drivers (SHAP), model health

Reduced budget → lower service level on low-margin families · Error costs change → move the quantile · Drift → PSI + monthly refit · Fairness → group audit, bias correction · Missouri tax rises → border volume recovers (H1 bounds it) · Licensing wave → plan net of ~60% diversion